

Mahsa Payami

U.S. Permanent Resident | payami.career@gmail.com | [LinkedIn](#) | [Google Scholar](#) | [Website](#)

Professional Summary

Atmospheric scientist and applied machine-learning researcher with a Ph.D. and current Georgia Tech M.S. training. Translates open-ended scientific questions into validated PyTorch models, reproducible experiments, and clear technical results. Built deep-learning emulators and data pipelines for terabyte-scale spatiotemporal datasets, led a five-person research team, and collaborated with NASA-funded partners.

Education

Georgia Institute of Technology, USA

Master of Science in Computer Science (ML specialization)

May 2027 (expected)

University of Houston, USA

Doctor of Philosophy in Atmospheric Sciences (*UH Presidential Fellowship*)

June 2025

Amirkabir University of Technology, Iran

Master of Science in Environmental Engineering

Jan 2019

Sharif University of Technology, Iran

Bachelor of Science in Civil Engineering

May 2016

Technical Skills

Programming and Data: Python, SQL, Pandas, NumPy, xarray, relational data, ETL pipelines

Machine Learning and Analysis: scikit-learn, TensorFlow, PyTorch, supervised and statistical ML, CNN, U-Net, model validation, sensitivity analysis, SHAP

Tools and Visualization: Git, Weights & Biases, Tableau, Matplotlib

Research Experience

UH Air Quality Forecasting and Machine Learning Group, University of Houston

2022 – 2025

- Led applied machine-learning research using large-scale atmospheric and emissions datasets, coordinating a five-member team's responsibilities, timelines and deliverables.
- Developed reproducible workflows for data ingestion, preprocessing, validation, experiment tracking, and scientific interpretation using Python, xarray, Pandas and Weights & Biases.
- Collaborated with NASA-funded research partners to evaluate modeling approaches and communicate findings to interdisciplinary scientific audiences. Co-authored five peer-reviewed publications and presented research at national and international conferences.

Selected Applied Research

Deep Learning-based Emulator (NASA-funded) | PyTorch, 1D/2D CNN, U-Net, SHAP – [Papers: [1](#), [2](#) & [3](#)]

- Built 1D/2D CNN and U-Net models in PyTorch to emulate the CMAQ air quality model, replacing expensive numerical solvers for nationwide air quality prediction.
- Achieved 95% accuracy and 1,064× GPU speedup over CPU-bound Fortran solvers; applied SHAP to identify NOx emissions and solar radiation as top prediction drivers.

Large-Scale Data Integration & Scenario Modeling (CRC-funded) | Python, Pandas, NumPy – [[Paper](#)]

- Built Pandas/SQL pipelines to integrate multi-source transportation, energy, and emissions data (CSV, relational tables) across four U.S. cities for scenario modeling.
- Projected city-level greenhouse gas and air quality impacts across multiple EV adoption and electricity generation scenarios; performed sensitivity analyses to quantify uncertainty.

Contrail Detection (OpenContrails) | PyTorch, U-Net, Diffusion Models, Satellite Imagery [[GitHub](#)]

- Developed a U-Net segmentation pipeline with diffusion-based augmentation for pixel-level contrail detection in multi-frame infrared satellite imagery. Engineered frame-difference temporal features and geometric and Fourier-domain priors to address extreme class imbalance, with roughly 1% positive pixels.
- Benchmarked multiple model and training variants across 20,500+ labeled samples using PR-AUC, Dice score, and skeleton recall.

Notable Awards

- Outstanding Graduate Researcher, University of Houston 2024 & 2025
- Best Visiting Team, Rice University's Datathon (Chevron Challenge) 2024